

MA 301
Test 1
Fall - 2014

Problem	points	out of
1		7
2		7
3		7
4		4
5		7
6		8
7		2
EC		5
Score		42

Name: _____

HR: _____

Signature: _____

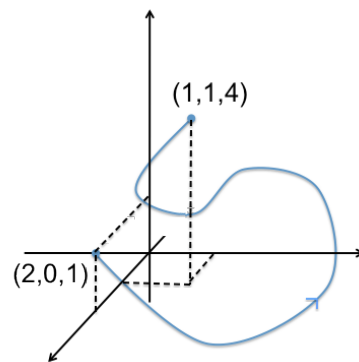
READ BEFORE YOU START
* Show all work. * Change your calculator mode to RADIANS. * Sloppy solutions may be penalized. * Use calculators ONLY for computations. * You must write up every problem in a clear and organized way. * If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Let $\mathbf{F} = [z^2, x - y, 7]$ and C be the intersection of the surfaces $x = y + z$ and $y = z^2$ with $0 \leq y \leq 9$.

(a) Compute $\int_C \mathbf{F} \cdot d\mathbf{r}$

(b) Completely set-up, but DO NOT compute the integral needed to find the length of C . (arclength $= \int_C 1 \, ds$)

Problem 2: Let $\mathbf{F} = \left[y - \ln(y), -\frac{x}{y} + x, 1 \right]$ compute the work required to move an object along C (shown below). Justify your work.



Problem 3: Let $\mathbf{F}(x, y) = (xe^y - 3xy)\mathbf{i} + \frac{3}{2}y^2\mathbf{j}$ and suppose C consist of the following 3 curves: the line segment from $(0,0)$ to $(2,0)$ then the curve $y = 4 - x^2$ from $(2,0)$ to $(0,4)$ and the line segment from $(0,4)$ to $(0,0)$.

(a) Sketch C

(b) Compute $\int_C \mathbf{F} \cdot d\mathbf{r}$

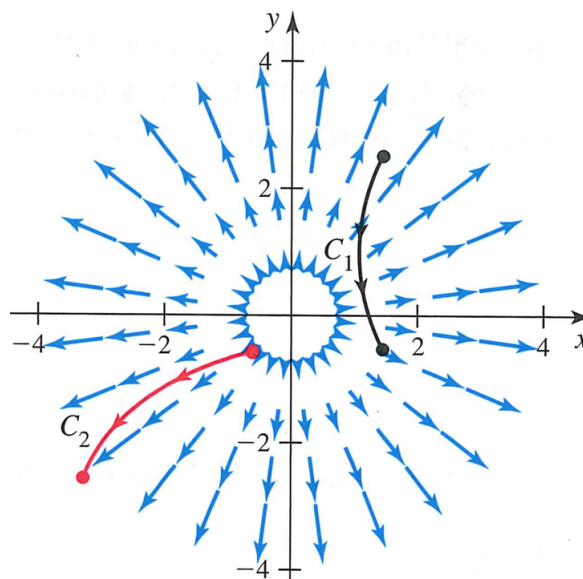
Problem 4: Suppose a straight wire from initial point $(0, 1, 5)$ to end point $(2, 2, 3)$ has a linear density function given by $\delta(x, y, z) = x^3z$. Find the total mass of the wire.

Problem 5: Answer the following true/false questions

- (a) True or False: If C is an ellipse and $\mathbf{F}$ is irrotational then $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$
- (b) True or False: If $\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} \neq 0$ for some closed path C , then $\mathbf{F}$ is conservative.
- (c) True or False: If $\mathbf{F} = \nabla f$ then $\mathbf{F}$ is conservative.
- (d) True or False: If $\mathbf{F}$ is conservative then the work required to move an object along any path is zero.
- (e) True or False: A vector field, $\mathbf{F}$, can be both incompressible and irrotational.
- (f) True or False: If $\mathbf{F}$ is path-independent and C is any closed curve, then $\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = 0$
- (g) True or False: Line integrals over all constant vector fields are path-independent.
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Problem 6: Let $\mathbf{F}$ be the vector field and C_1 and C_2 be the curves shown in the provided figure.

- (a) Is $\mathbf{F}$ incompressible?
- (b) Is $\mathbf{F}$ irrotational?
- (c) Is $\mathbf{F}$ conservative?
- (d) Would the work for a mass traveling along C_1 be positive, negative, or zero? Explain.
- (e) Would the work for a mass traveling along C_2 be positive, negative, or zero? Explain.
- (f) Which is larger: $|\text{curl}(\mathbf{F}(2,2))|$ or $\text{div}(\mathbf{F}(2,2))$? Explain.



Problem 7: Let $f(x, y, z)$ be a scalar-valued function. Using the concepts from this class, explain why $\text{curl}(\nabla f) = \mathbf{0}$ no matter what $f(x, y, z)$ is. If you cannot explain it, then prove it mathematically.

Extra Credit: Let C be a simple closed curve enclosing a region R .

(a) Use Green's Theorem to show $\oint_C x dy = \text{Area}(R)$

(b) Use this to find the area of the region with boundary given by $\mathbf{r}(t) = [\sin 2t, 2 \sin t]$ with $0 \leq t \leq \pi$.

Hint: The trig identity: $\sin(2t) = 2 \sin t \cos t$ might come in handy.

